

CCP6224 Object Oriented Analysis and Design

Lab 6 – Structural Design Patterns

Lab outcomes

By the end of today's lab, you should be able to

- differentiate structural design patterns and apply them to particular scenarios

Design patterns

- Map the reasons of when to implement with the correct matching design pattern

DP

Adapter •

Façade •

Composite •

Bridge •

Reasons to implement

- hides the complexities of the system and provides an interface to the client using which the client can access the system
- to create complex objects from individual parts in certain order/algorithm
- adapts existing legacy code
- to create duplicate objects without affecting application performance
- separate an object from underlying operations so that any existing operation can be replace later without affecting it (e.g. separating GUI from backend operations)
- represent multiple related objects (usually a hierarchy) with a generalized object such that each object and composition of objects are treated uniformly (i.e. same interface)
- used when the operating behaviour or algorithm of a class needs to be changed during runtime

Note: Most of the following questions are past Year Exam Questions

For each scenario, what design pattern would you use? Explain why you would choose this design pattern.

- Henry is a software engineer working on a software project that has to interface with hardware from multiple independent vendors. Although each vendor has provided specific libraries and pre-compiled class objects, he needs all the components to work together in the new system

Design pattern:

Reasoning:

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3. You have a complex but useful subsystem to complete your task. You have many coders working under you who are fresh graduates. They will need to code different applications in the system that will use some of the functionality of the above subsystem, but you are concerned that they will get confused or overwhelmed by the complexity of that subsystem.

Design pattern:
Reasoning:

4. The system you are working on includes, among other things, a catalogue of millions of photos. These photos are stored in repository with metadata. There is not enough RAM to load all these photos into memory, but you need to be able to access them whenever the user or system processes wants it based on their metadata choices.

Design pattern:
Reasoning:

5. You have an existing automation control system which is being used to control most of the machines in a factory. The factory has several machines which have their own API which do not fit in the abovementioned control system. What design pattern would you use to enable the existing automation control system to control those machines?

Design pattern:
Reasoning: